# Secure Federated Learning Protocol Sequence

Consortium identity, encrypted updates, threshold decryption and auditable model release

CA Consortium CA
H Hospitals A...N
B Permissioned Ledger
S Aggregation Server
K Threshold Key Committee

## 01 One-time setup and periodic key rotation

1. Register institution and authorized role
2. Issue consortium certificate
3. Participate in distributed key generation
4. Collective CKKS public key
5. Protected t-of-n secret-key shares

## 02 Federated learning round r

6. Open signed round with model hash and nonce
7. Publish verified round metadata
8. Broadcast signed global model
9. Verify hash, consent and model version
10. Train, clip, add DP noise and compress
11. CKKS encrypt and sign update
12. Ciphertext + signature + nonce + round ID

## 03 Server-side admission and aggregation

13. Verify identity, signature and freshness

Invalid identity, replay or policy failure

14. Record rejection evidence
15. Reject update

Else: valid update

16. Store ciphertext digest and acceptance status
17. Homomorphic weighted aggregation

## 04 Threshold decryption and dropout handling

Quorum reached despite client dropout

18. Submit aggregate ciphertext only
19. Collect at least t authorized key shares
20. Decrypted aggregate, never individual updates
21. Validate accuracy, drift and anomaly indicators

Else: quorum not reached

22. Mark round incomplete
23. Retry without exposing submitted updates

## 05 Candidate validation and global-model decision

Candidate passes validation

24. Commit global-model hash and reputation updates
25. Securely broadcast accepted global model

Else: candidate fails validation

26. Record rollback and quarantine decision
27. Restore last trusted global model

Solid arrow: request/action  Dashed arrow: verified response or secure broadcast  Numbering follows protocol execution order.